

■ Day 4 — Compression Encoding, VACUUM & ANALYZE

■ Topics Covered

- Compression Encoding
 - AUTO Encoding Internals
 - Manual Encoding Scenarios
 - VACUUM Internals
 - ANALYZE Internals
 - Query Planner Statistics
 - Production Warehouse Maintenance
 - Hands-On Labs
-

■ Compression Encoding

Redshift compresses columns separately because it is columnar.

Benefits:

- Faster scans
- Less storage
- Reduced I/O
- Better performance

■ Encoding Types

AZ64 → Numeric columns

LZO → VARCHAR columns

BYTEDICT → Low-cardinality values

DELTA → Sequential timestamps/numbers

RAW → No compression

■ AUTO Encoding Internals

During COPY Redshift:

- Samples data blocks
- Detects patterns
- Checks repetition/cardinality
- Estimates compression ratio
- Assigns encoding automatically

■ LAB 1 — Create Optimized Table

```
CREATE TABLE optimized_transactions (  
    transaction_id BIGINT ENCODE AZ64,  
    customer_id BIGINT ENCODE AZ64,  
    merchant_id BIGINT ENCODE AZ64,  
    amount DECIMAL(10,2) ENCODE AZ64,  
    payment_mode VARCHAR(20) ENCODE LZO,  
    status VARCHAR(20) ENCODE BYTEDICT,  
    transaction_time TIMESTAMP ENCODE DELTA  
)
```

```
DISTSTYLE KEY
DISTKEY(customer_id)
SORTKEY(transaction_time);
```

■ LAB 2 — COPY Data from S3

```
COPY optimized_transactions
FROM 's3://redshift-demo-bucket/transactions.csv'
IAM_ROLE 'YOUR-IAM-ROLE-ARN'
CSV
IGNOREHEADER 1;
```

■ LAB 3 — Query Data

```
SELECT *
FROM optimized_transactions;
```

■ VACUUM Internals

DELETE operation only marks rows for deletion.

Without VACUUM:

- wasted storage
- unsorted blocks
- slower scans

```
DELETE FROM optimized_transactions
WHERE payment_mode='WALLET';

VACUUM optimized_transactions;
```

■ ANALYZE Internals

ANALYZE collects statistics for query planner.

Statistics include:

- row counts
- cardinality
- value distribution

```
ANALYZE optimized_transactions;

ANALYZE COMPRESSION optimized_transactions;
```

■ Real Production Scenarios

Scenario 1 — Skewed Sample

Initial COPY sample may not represent future data.

Scenario 2 — Huge VARCHAR Growth

Small strings later become JSON payloads.

Scenario 3 — Sequential Timestamps

Manual DELTA encoding may outperform AUTO.

Scenario 4 — Petabyte Warehouses

Even 5% compression improvement saves huge cost.

■ Engineer Monitoring Metrics

- Query latency
- Scan bytes
- CPU usage
- Storage growth
- Block reads
- Data skew

■ Day 4 Outcome

- Learned Compression Encoding
- Learned AUTO Encoding
- Practiced VACUUM
- Practiced ANALYZE
- Understood Production Optimization